

Building Sustainability Index (BASIX)

(An initiative of the New South Wales Government, Australia)

Content

- What is BASIX?
- Key areas in BASIX
- BASIX Requirements
- BASIX Certificate
- Increase to BASIX standards: New BASIX Standards
 - Comparison

What is **BASIX**?

- an initiative by the **New South Wales Government** in **Australia** aimed at **ensuring residential buildings meet certain sustainability targets**
- a **sustainability assessment tool**
- sets **minimum standards** for **water, thermal comfort, and energy efficiency**

Key Areas assessed in **BASIX**

Water Efficiency	Energy Efficiency	Thermal Comfort
• Measure to reduce water consumption • Ex: rainwater tanks, water-efficient fixtures, landscaping with drought-resistant plants	• Measure to reduce energy consumption and greenhouse gas emissions • As: efficient lighting, insulation, solar panels, high-performance windows	• Design feature that enhance the comfort of building occupants • Reducing need for artificial heating & cooling

BASIX requirements

Water Efficiency	Energy Efficiency	Thermal Comfort
Installation of water-efficient fixtures and fittings , such as low-flow showerheads, taps, and dual-flush toilets.	Use of energy-efficient lighting, appliances, and heating/cooling systems .	Proper orientation of the building to maximize natural light and heat during winter and minimize it during summer.
Use of rainwater tanks for outdoor use, toilet flushing, and laundry	Installation of insulation in walls, ceilings, and floors to reduce energy consumption	Use of appropriate glazing, shading devices, and thermal mass to maintain stable indoor temperatures.
Incorporation of drought-tolerant landscaping and efficient irrigation systems .	Incorporation of solar panels or other renewable energy sources	Ensuring that the building design reduces reliance on mechanical heating and cooling
	Design considerations for natural ventilation, daylighting, and passive solar heating .	

About BASIX Certificate

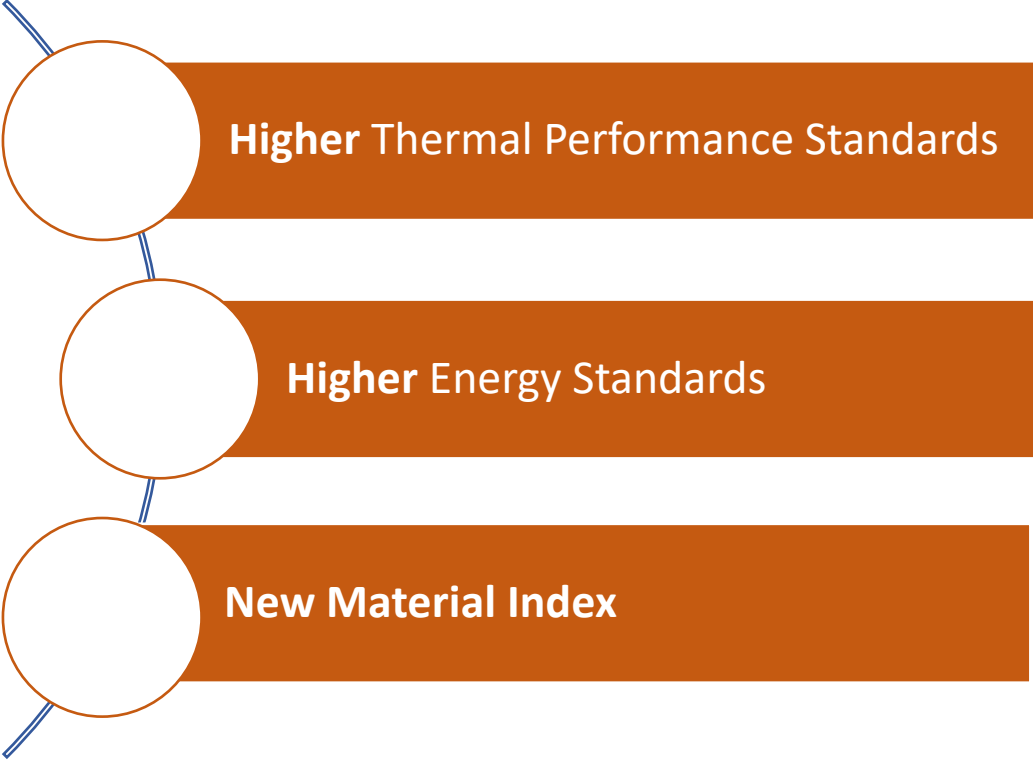
Aspects	Description
Cost of Obtaining	Depends on complexity and size of the project , and whether the resident choose to have the NatHERS thermal assessment included
Validity	3 months before submitted to the council or an accredited certifier After lodging, valid for entire duration of the Development Application
Is it do it yourself ?	Yes

NatHERS ?

- Nationwide House Energy Rating Scheme
- A **national scheme** provides **energy ratings** that can be **used for the thermal comfort component of BASIX** through the simulation method
- It is **mandatory** in **NSW**
- NatHERS standards **provide minimum requirements for health and safety in the design and construction of new homes** in Australia

Increase to BASIX Standards

(came into effect on 1 October 2023)



Higher Thermal Performance Standards

Higher Energy Standards

New Material Index

“New BASIX standards will help build **more comfortable homes, cut energy costs, and contribute to our **target of net zero homes** (few carbon emissions) **by 2050”****

Changes to Energy Standards

Building Type	Current Energy Standards	Higher Energy Standards
Detached houses	Standards do not vary with floor area	- Standards will vary with the total floor area. - To account for energy consumption and the number of people that live at property
High-rise apartment building	Same standards apply to apartment buildings 6 storeys and higher	Separate standards: 6- 20 storeys 21 storeys and higher Account for the energy consumption of shared services specified in high-rise buildings

Source: NSW Government website

Changes to Thermal Performance Standards

Building Type	Current Standards	Higher Standards
Detached or attached houses	Heating and cooling caps to deliver average 5.5 to 6 NatHERS star outcome	• 7 NatHERS stars • Revised heating and cooling caps
Apartment	• Heating and cooling caps to deliver average 5.5 to 6 NatHERS star outcome • Maximum heating and cooling caps for individual apartments	• Across the buildings: average 7 NatHERS stars average heating and cooling caps • Individual apartments: minimum 6 NatHERS stars maximum heating and cooling caps

Source: NSW Government website

New Material Index

- It **calculates and reports on the embodied emissions of a home**
- It **estimates the volume of different materials used in construction and applies the emission factors** for the materials, which represent embodied emissions from the production of each building material

References:

<https://www.ecosuresolutions.com.au/faq>

<https://www.planningportal.nsw.gov.au/BASIX-standards#higher-thermal-performance-standards>